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How to
list

What is it For?

Your lab report is your means of communicating to the world what you have observed.

You are writing to an audience who has never seen your experiment.

You should make clear descriptions; use writing to ‘put them in the room.’ All writing should be objective.

What is it For?

You are part of a scientific community, and the people who participate in this community tend to share the same values

You have a responsibility to the rest of the scientific world to report your findings

The people reading such reports have two primary goals:

to gather the information presented

to know that the findings are legitimate

Your job as a writer, then, is to fulfill these two goals.

Preparing the Lab Report

The best way to prepare to write the lab report is to make sure that you fully understand everything you need to about the experiment. Obviously, if you don't quite know what went on during the lab, you're going to find it difficult to explain the lab satisfactorily to someone else. **To make sure you know enough to write the report, complete the following steps:**

Steps of the Lab Report

Title / Date / Lab Partners

Introduction

Purpose/Objective

Hypothesis

Materials

Procedure

Observations & Data

Conclusion



Title / Date / Lab Partners

Title should clearly state what relationship is being investigated (the variables)

Introduction

Should be a general background of the topic, not too specific



Purpose/Objective

What are you going to learn? What is the reason for doing the experiment?

What do you expect to gain through the experiment?

This section should be specific topics relating to your lab

Never use the word “prove.” Better terms are: test, verify, investigate, confirm, support.

Hypothesis

If-then statement

What you expect the results to be

Clearly states how the variable will affect the outcome

It is testable

Identify your variables

Materials

List all materials used in experiment

Procedure

Step-by-step experiment (1 ,2 ,3, etc.)

Be efficient (repeat similar steps at the same time)

Enough detail that someone else could replicate it –
not you, not someone in your group, not your
teacher, ...some random person

Identify what you are measuring, what units to use,
how to measure it and where to record your data

Observations & Data

Qualitative (narrative / “words”) and quantitative (numerical / “numbers”) observations

Tables

Diagrams

Graphs: graphs must include title, labeled axes, including scale and units, and best-fit trendline (if appropriate) –CHOOSE THE CORRECT TYPE OF GRAPH!

Calculations (shown, including equations used)

Conclusion

Summarize the experiment: what is the relationship/big picture that was discovered?

Must state whether the experiment “supports” or “disproves” the hypothesis

Acknowledge any anomalous data, or deviations from what you expected

Must include possible sources of experimental error, calculation of percent error

The implications of all graphed data should be discussed - what does the graph show you?

Suggestions for improving the experiment in the future - HOW you would fix it

Oreo Lab Report

In teams of three or four, you will use the scientific process to design and carry out a valid/reliable experiment to test:

Do “Double Stuff Oreos” actually have twice the stuffing as regular Oreos?

**BREAK INTO GROUPS
NOW**

Oreo Lab Report

One person from your group is to open the lab report template and share it with the members of your group, making sure everybody has the ability to edit

Oreo Lab Report - Hypothesis

Does your group think there is twice as much icing in a Double Stuf (DS) Oreo as compared to a regular Oreo?

Does your group think there is less than twice as much icing in a Double Stuf (DS) Oreo as compared to a regular Oreo?

Does your group think there is more than twice as much icing in a Double Stuf (DS) Oreo as compared to a regular Oreo?

Does your group think there is the same amount of icing in a regular and DS Oreo?

If a DS Oreo has _____ icing as a regular Oreo
then the mass of the icing of a DS Oreo will be
_____ than the mass of a regular Oreo.

Oreo Lab Report - Materials

Under “Materials” section, write the following:

Triple Beam Balance

3 Regular Oreos

3 Double Stuff Oreos

1 plastic knife

1 paper plate

Oreo Lab Report - Observations

You are going to need to create some data tables to record your measurements and calculations

If you are finding the mass of the cookies, the mass of the wafers and then using them to find the mass of the icing, you will need 3 data tables.

Follow the steps on the next 3 slides to create your data tables under the observations and data section of your lab report.

Oreo Lab Report - Data Tables

Create a 3 column by 5 row Data Table in “Observations” section to record the Oreo mass measurement

Merge the top three cells

Label the cells as follows:

Measured Mass of Oreos		
Oreo #	Regular Oreo (g)	Double Stuff Oreo (g)
1		
2		
3		

Oreo Lab Report - Data Tables

Create a 3 column by 5 row Data Table in “Observations” section to record the Oreo wafer mass measurement

Merge the top three cells

Label the cells as follows:

Measured Mass of Oreo Wafers		
Oreo #	Regular Oreo (g)	Double Stuff Oreo (g)
1		
2		
3		

Oreo Lab Report - Data Tables

Create a 3 column by 6 row Data Table in “Observations” section to record the Oreo icing mass calculations

Merge the top three cells

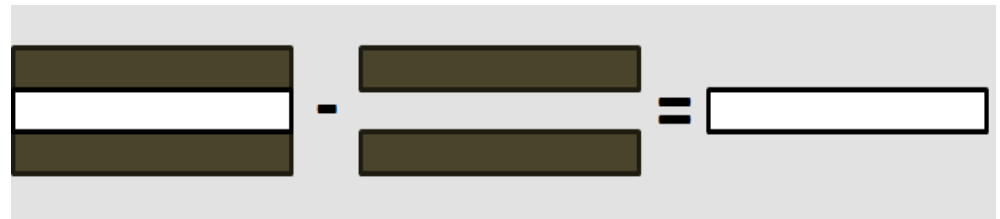
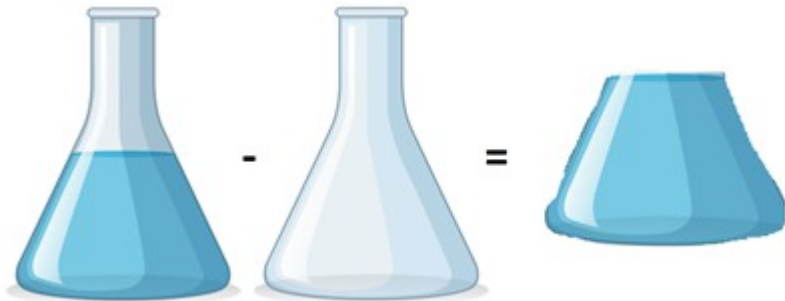
Label the cells as follows:

Calculated Mass of Oreo Icing		
Oreo #	Oreo #	Oreo #
1	1	1
2	2	2
3	3	3
Average		

Oreo Lab Report - Procedures

As a group develop a ***specific*** set of procedures you will use to determine if “Double Stuff Oreos” actually have twice the stuffing as regular Oreos?

Remember how we measured the mass of water? Can you use that method here?



Oreo Lab Report

Complete the following sections as a group prior to coming to next class where you will carry out the experiment:

Title / Date / Lab Partners

Introduction

Purpose/Objective

Hypothesis

Materials

Procedure